

Buy-and-Hold vs. Market Timing: The Consequences of Portfolio Turnover

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Following the financial crisis and Great Recession of the 2008-2009 period, several financial publications have declared that “the era of buy-and-hold is dead.” This pronouncement has been made not only by those in the media, but by countless analysts, strategists and portfolio managers as well. The general sentiment is that those who buy shares of businesses and hold them for the long run are on a fool’s errand—that large market swings will inevitably destroy any meaningful return over time. After all, the S&P 500 had a negative 1% annual rate of return over the decade ended in December 2009. The conclusion is that in order to generate a meaningful return, one *must* time the market’s upswings and downswings, and better yet, time the short-term ups and downs of individual stocks as well.

Clients and other friends of Horizon are certainly aware that we eschew this line of thinking; an overview of our long-term investment philosophy—buying businesses with long product lifecycles and profiting from the equity yield curve—need not be repeated here. Instead, let us examine this topic from a different angle. Let us presume that investors can, in fact, time the market’s swings with outstanding precision. Assume investors have the prescience to know when the market has reached a “top” and when the market has reached a “bottom”, and buy and sell accordingly. Further assume they have mastered this practice to such a degree that they can generate a pre-tax rate of return of 20% per annum over a period of decades—a feat achieved perhaps by only a handful of investors in history. With the constant trading and portfolio turnover required to carry out this strategy, how would these investors fare on an after-tax basis over 50 years?

To answer this question, let us first examine how an investor would fare over 50 years without trading (i.e., the buy-and-hold strategy). Consider a young individual, fresh out of college at age 22, who has amassed a sum of \$1,000. This person sets about investing that sum of money for the next half-century, until age 72, and earns an annual rate of return of 20%. At age 72, the individual sells all the investments with the notion of enjoying the benefits of deferred consumption and skillful investing. Under this set of circumstances, the sum of money available in 50 years would be \$9,100,438. At that time, assuming applicable taxes of 28%, the after-tax sum of money available for expenditure would be \$6,552,315.

Table 1: Growth of \$1,000 at 20% per annum for 50 years

<i>Year</i>		<i>Year</i>	
0.	\$ 1,000	26.	114,475
1.	1,200	27.	137,371
2.	1,440	28.	164,845
3.	1,728	29.	197,814
4.	2,074	30.	237,376
5.	2,488	31.	284,852
6.	2,986	32.	341,822
7.	3,583	33.	410,186
8.	4,300	34.	492,224
9.	5,160	35.	590,668
10.	6,192	36.	708,802
11.	7,430	37.	850,562
12.	8,916	38.	1,020,675
13.	10,699	39.	1,224,810
14.	12,839	40.	1,469,772
15.	15,407	41.	1,763,726
16.	18,488	42.	2,116,471
17.	22,186	43.	2,539,765
18.	26,623	44.	3,047,718
19.	31,948	45.	3,657,262
20.	38,338	46.	4,388,714
21.	46,005	47.	5,266,457
22.	55,206	48.	6,319,749
23.	66,247	49.	7,583,698
24.	79,497	50.	\$ 9,100,438
25.	95,396	28% tax:	<u>(2,548,123)</u>
		After-tax:	\$ 6,552,315

Now, refer back to the initial scenario. Let's assume the prescient market-timer turns over his or her portfolio of investments only once every five years. In reality, those who stress the importance of market timing would more likely buy and sell securities every few months, if not more frequently. This scenario also presumes that all of the investment return is from capital gains — that none of it is from dividends. This distinction is very important for the purposes of this calculation because, in the case of dividends, taxes are due in the year in which the dividends are realized. This exercise also assumes that there is an average 20% turnover per annum, but that all of the turnover occurs in five-year increments. In other words, the portfolio has no turnover until one day, at the end of the fifth year, there is 100% turnover in the portfolio, all the profits are realized, and taxes of 28% are paid. It is important to note that if the 20% turnover per annum were realized in a more linear manner, which would mean incurring the tax liability more frequently, the average principal sum received would be lower than it would be if the capital gains were deferred until the end of each five-year period. The following table illustrates this example.

Table 2: Growth of \$1,000 at 20% per annum, with 100% portfolio turnover every 5 years

<u>Year</u>	<u>Profit</u>	<u>Tax</u> <u>28%</u>	<u>Total</u> <u>(net of tax)</u>	<u>Year</u>	<u>Profit</u>	<u>Tax</u> <u>28%</u>	<u>Total</u> <u>(net of tax)</u>
0.	\$ 1,000			26.	45,783		
1.	1,200			27.	54,939		
2.	1,440			28.	65,927		
3.	1,728			29.	79,112		
4.	2,074			30.	94,935	56,783	15,899
5.	2,488	1,488	417	31.	94,843		79,036
6.	2,486			32.	113,811		
7.	2,983			33.	136,574		
8.	3,580			34.	163,888		
9.	4,296			35.	196,666	117,630	32,937
10.	5,155	3,083	863	36.	196,476		163,730
11.	5,150			37.	235,771		
12.	6,180			38.	282,925		
13.	7,416			39.	339,510		
14.	8,899			40.	407,412	243,682	68,231
15.	10,679	6,387	1,788	41.	407,017		339,181
16.	10,668			42.	488,420		
17.	12,802			43.	586,104		
18.	15,362			44.	703,325		
19.	18,435			45.	843,990	504,809	141,347
20.	22,122	13,231	3,705	46.	843,172		702,643
21.	22,100			47.	1,011,807		
22.	26,520			48.	1,214,168		
23.	31,824			49.	1,457,001		
24.	38,189			50.	1,748,402	1,045,758	292,812
25.	45,827	27,410	7,675				\$1,455,589
			38,152				

Recall that in the initial scenario, when taxes were paid only once, at the end of the 50-year period, the after-tax sum of money was \$6,552,315. In the second example, in which taxes are paid on gains every five years, the after-tax amount at the end of 50 years is just \$1,455,589. The difference amounts to a sum of money that is 78% less. As noted previously, had the gains been realized more frequently during the half-decade periods (e.g., 20% per year instead of 100% every five years), the amount available at the end of 50 years would be lower still.

Now, let us go a step further and assume our market-timer were to turn over the portfolio every year, which is the equivalent of 100% annual portfolio turnover (and more closely approximates reality). In this scenario, maintaining the 20% pre-tax rate of return, but with taxes paid on gains every year, the after-tax sum of money available in 50 years would be just \$834,260.

Table 3: Growth of \$1,000 at 20% per annum, with annual turnover of 100%

<u>Year</u>		<u>Profit</u>	<u>Tax</u> <u>28%</u>	<u>Total</u> <u>(net of tax)</u>	<u>Year</u>		<u>Profit</u>	<u>Tax</u> <u>28%</u>	<u>Total</u> <u>(net of tax)</u>
0.	\$ 1,000								
1.	1,200	200	56	1,144	26.	34,660	5,777	1,617	33,043
2.	1,373	229	64	1,309	27.	39,651	6,609	1,850	37,801
3.	1,570	262	73	1,497	28.	45,361	7,560	2,117	43,244
4.	1,797	299	84	1,713	29.	51,893	8,649	2,422	49,471
5.	2,055	343	96	1,959	30.	59,366	9,894	2,770	56,595
6.	2,351	392	110	2,242	31.	67,914	11,319	3,169	64,745
7.	2,690	448	126	2,564	32.	77,694	12,949	3,626	74,068
8.	3,077	513	144	2,934	33.	88,882	14,814	4,148	84,734
9.	3,520	587	164	3,356	34.	101,681	16,947	4,745	96,936
10.	4,027	671	188	3,839	35.	116,323	19,387	5,428	110,895
11.	4,607	768	215	4,392	36.	133,074	22,179	6,210	126,864
12.	5,271	878	246	5,025	37.	152,236	25,373	7,104	145,132
13.	6,030	1,005	281	5,748	38.	174,158	29,026	8,127	166,031
14.	6,898	1,150	322	6,576	39.	199,237	33,206	9,298	189,939
15.	7,891	1,315	368	7,523	40.	227,927	37,988	10,637	217,291
16.	9,028	1,505	421	8,606	41.	260,749	43,458	12,168	248,580
17.	10,328	1,721	482	9,846	42.	298,297	49,716	13,921	284,376
18.	11,815	1,969	551	11,263	43.	341,251	56,875	15,925	325,326
19.	13,516	2,253	631	12,885	44.	390,391	65,065	18,218	372,173
20.	15,462	2,577	722	14,741	45.	446,608	74,435	20,842	425,766
21.	17,689	2,948	825	16,863	46.	510,919	85,153	23,843	487,076
22.	20,236	3,373	944	19,292	47.	584,492	97,415	27,276	557,215
23.	23,150	3,858	1,080	22,070	48.	668,659	111,443	31,204	637,454
24.	26,484	4,414	1,236	25,248	49.	764,945	127,491	35,697	729,248
25.	30,297	5,050	1,414	28,884	50.	875,098	145,850	40,838	\$ 834,260

These three examples demonstrate the tremendous destruction of profit that taxes impose on a portfolio when one trades more frequently. Even given the same pre-tax rate of return of 20%, the sum

of money available for consumption in 50 years declines from \$6,552,315 with no trading to just \$834,260 with annual turnover—a decline of 87%. The market-timer with annual turnover would need to generate a pre-tax rate of return of 26.7% to match the after-tax proceeds of the buy-and-hold investor with a rate of return of 20%. Viewed in a different manner, to generate after-tax proceeds of \$834,260, the buy-and-hold investor would only need to generate a pre-tax rate of return of 15.2%—a still remarkable but no doubt more realistic pursuit.

Furthermore, it bears consideration that in all three examples, the same tax rate of 28% was applied. This was chosen to approximate the historical level of long-term capital gains, plus taxes assessed at the state level. However, if a market-timer were to trade in and out of stocks at periods shorter than a year, a short-term capital gains tax would be applied at a significantly higher rate, and therefore the profit destruction over 50 years would be far greater. It might also be worth considering, given the mounting fiscal problems at the federal and state levels, in which direction tax rates in the U.S. are likely to go in the future.

Of course, there are additional costs of trading not taken into account in these examples. For example, this exercise assumes no sales commissions or the market impact of trading (i.e., slippage). For less liquid securities, there is also the considerable cost of the bid/ask spread on trades. To demonstrate the destructive impact of these other costs of trading, let us assume they add up to a 50-basis points (i.e., one-half of one percent) per annum decrease in the overall rate of return. Whereas \$1,000 invested at a 20% rate of return over 50 years results in pre-tax profits of \$9,100,438, the same amount invested at a 19.5% rate of return over 50 years results in pre-tax profits of just \$7,385,761—a difference of \$1,714,677.

There may be investors who have constructed profitable trading strategies that call for high turnover, moving from sector to sector or company to company to earn high rates of return. Certainly there are portfolio managers who exhibit high turnover and have outperformed the market over long periods of time. If one were to study the great stock market investors in history, one would find that different people have devised different and clever ways to amass wealth. However, what is beyond dispute is that investors, by trading more frequently, create for themselves a higher requirement for success to match the after-tax performance of long-term horizon investors.

Addendum

The following table provides figures for actual mutual fund turnover for different categories of funds, which ranged from 77% to 88% in 2010.

<u>Mutual Fund Type</u>	<u>Average Annual Turnover *</u>
U.S. Large-Cap	80%
U.S. Mid-Cap	87%
U.S. Small-Cap	88%
International	77%

** Figures are estimates for 2010. Sourced from Bloomberg.*

These statistics are not widely commented upon in the press or by investment management companies themselves. Instead, the overwhelming focus is on the pre-tax annual returns of funds, while the focus should arguably be on after-tax returns.

For example, consider a benchmark return of 10% over a certain timeframe (a benchmark being the S&P 500 or any other widely used index). Asset Manager "A", who reports an annualized return of 10.5% (i.e., slight outperformance) with a turnover ratio of 100% will likely attract a large sum of assets. Meanwhile, Fund Manager "B", who reports a return of 9.5% (so-called underperformance, on a pre-tax basis) with a turnover ratio of 20% over that timeframe might be subject to a large-scale withdrawal of assets. A scenario such as this is not uncommon in the investment management industry. However, such performance would result in Manager "B" actually producing a superior after-tax return. Manager "A" would, for the most part, incur taxes at the short-term capital gains rate. On the other hand, with a turnover ratio of just 20%, Manager "B" would incur taxes at the long-term capital gains rate. One should also consider taxes assessed at the state level, but since rates vary by state of residence, these are ignored in this example.

Growth of \$1,000 Invested with 2 Different Managers Over 30 Years

	<u>Manager "A"</u>	<u>Manager "B"</u>
Annualized return	10.5%	9.5%
Average annual turnover	100%	20%
Tax rate *	35%	20%
Amount available, <i>before</i> taxes, in 30 years	\$ 19,993	\$ 15,220
Amount available, <i>after</i> taxes, in 30 years	\$ 7,247	\$ 9,661
After-tax return	6.8%	7.9%

* Manager "A" incurs the short-term capital gains rate (equal to the Federal ordinary income tax rate in 2011 for the highest tax bracket), while Manager "B" incurs a long-term capital gains rate of 20% (while the long-term rate is currently 15%, this is set to increase to 20% after 2012).

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